

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B.Tech. (CL/EC/ME/ CE/EE/IT)

SEM - II University Examination, 2010-11

MA - 102 Engineering Mathematics-II

Date: 16.12.11, Friday

Time: 1.30 p. m to 4.30 p. m.

Maximum Marks: 70

Instructions: (1) All questions are compulsory.

(2) Indicate clearly, the option you attempt along with its respective question number.

(3) Figures to right indicate marks.

(4) Rough work is done in the last page of main answer book, don't write anything on the question paper.

SECTION-I

Q-1 (a) Solve by the method of variation of parameters : $\frac{d^2y}{dx^2} + a^2y = \sec ax$. [05]

(b) Solve : $(x^2 - 4xy - 2y^2)dx + (y^2 - 4xy - 2x^2)dy = 0$. [03]

(c) Solve : (i) $\sqrt{p} + \sqrt{q} = 1$ (ii) $p + q = pq$ [03]

Q-2 (a) Solve : $\frac{dy}{dx} + \frac{4x}{x^2 + 1}y = \frac{1}{(x^2 + 1)^3}$. [04]

(b) Solve: (i) $y''' + 6y'' + 11y' + 6y = 0$ (ii) $(D^2 + 2kD + k^2)y = 0$. [04]

(c) Solve the initial-value problem: $y'' + 2y' + y = e^{2x}$, $y(0) = -1$, $y'(0) = 1$. [04]

OR

Q-2 (a) Solve: $\frac{dy}{dx} - 2y \tan x = y^2 + \tan^2 x$. [04]

(b) Solve: $y'' + 5y' + 6y = x^2 + 3x$. [04]

(c) Solve : $x^2y'' + xy' + y = x^2 + 1$. [04]

Q-3 (a) Solve $(mz - ny)\frac{\partial z}{\partial x} + (nx - lz)\frac{\partial z}{\partial y} = ly - mx$. [04]

(b) Solve $p(1 + q) = qz$ [03]

(c) Solve (i) $p^2 + q^2 = 1$ (ii) $p - x^2 = q + y^2$ (iii) $z = px + qy + 2\sqrt{pq}$ [05]

OR

Q-3 (a) Solve $y^2p - xyq = x(z - 2y)$. [04]

(b) Solve $p^2(1 + q^2) = q(z - a)$. [04]

(c) Solve (i) $pq = 1$ (ii) $z = p^2 + q^2$. [04]

SECTION-II

Q-4 (a) Find all the Eigen values and Eigenvectors of the matrix $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$ [05]

(b) Prove $J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$. [03]

(c) Evaluate $\iint_R xy \, dy \, dx$; R is the positive quadrant of the circle $x^2 + y^2 = a^2$ [03]

Q-5 (a) If $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$, Prove that $A^5 - 3A^4 + A^3 - 7A^2 + 5A + I = 61I - 43A$ [04]

(b) Prove that $\int_0^1 \int_0^x \int_0^{\sqrt{x+y}} z \, dx \, dy \, dz = \frac{1}{4}$. [04]

(c) Find the area between the parabola $y^2 = 4ax$ and $x^2 = 4ay$. [04]

OR

Q-5 (a) Change the order of integration in the integral $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} \, dy \, dx$ and evaluate it. [04]

(b) If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, compute $A^2 - 4A - 5I$ and A^{-1} . [04]

(c) Prove (i) $P_n(1) = 1$ (ii) $P_n(-x) = (-1)^n P_n(x)$. [04]

Q-6 (a) Show that $\operatorname{erf}(-x) = -\operatorname{erf}(x)$ [03]

(b) If $B = \begin{bmatrix} -1 & 2+i & 5-3i \\ 2-i & 7 & 5i \\ 5+3i & -5i & 2 \end{bmatrix}$, show that B is Hermitian matrix and iB is a [05]

Skew-Hermitian matrix.

(c) Evaluate $\int_0^\infty x^2 e^{-x^4} \, dx$. [04]

OR

Q-6 (a) Express $f(x) = 4x^3 - 3x^2 + 2x + 1$ in terms of Legendre polynomials. [04]

(b) Evaluate (i) $\int_0^\infty e^{-x^3} \, dx$ (ii) $B(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta \, d\theta$ [04]

(c) By changing into polar co-ordinates, evaluate the integral $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} \, dx \, dy$. [04]

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